Proof for Theorem 1 (Thm:coverage-Algorithm-Known-Delta; Hul C Finite-Sample Coverage And Exact Coverage Under The Stated Additional Conditions)

Generated by Statopia Prover

October 31, 2026

Statement provenance. Additional conditions were adopted automatically without human confirmation. This report describes the proof under the recorded conditions.

Key Lemmas

Lemma 1 (Measurability and independence factorization of the noncoverage event)**.** *Let $(\Omega, \mathcal{F}, P)$ be a probability space, let $\theta_0 \in \mathbb{R}$, let $\widehat{\theta}_j : \Omega \to \mathbb{R}$ for $j \geq 1$, let $U : \Omega \to \mathbb{R}$, let $B, k \in \mathbb{N}$, and let $\tau \in \mathbb{R}$. Write*

$$\mathcal{G}_B = \bigvee_{j=1}^{B} \sigma(\widehat{\theta}_j)$$

for the σ-algebra generated by the first B estimators. Assume:

(i) each $\widehat{\theta}_j$ is Borel measurable, and U is Borel measurable;

(ii) $B \geq 1$;

(iii) $1 \leq k \leq B$;

(iv) $\sigma(U)$ and $\mathcal{G}_B$ are independent under P.

Define the noncoverage event

$$N_k = \Big(\{\omega : \exists\, j \in \{1, \ldots, k\} \text{ with } \widehat{\theta}_j(\omega) \leq \theta_0\} \cap \{\omega : \exists\, j \in \{1, \ldots, k\} \text{ with } \widehat{\theta}_j(\omega) \geq \theta_0\}\Big)^c,$$

the event that θ_0 is not in the closed interval between the minimum and maximum of $\widehat{\theta}_1(\omega), \ldots, \widehat{\theta}_k(\omega)$. Then

$$N_k \in \mathcal{F}, \qquad N_k \in \mathcal{G}_B,$$

and

$$P\big(N_k \cap \{U \le \tau\}\big) = P(N_k)\, P(U \le \tau), \qquad P\big(N_k \cap \{\tau < U\}\big) = P(N_k)\, P(\tau < U).$$

Proof. Put $\mathcal{G} = \mathcal{G}_B = \bigvee_{j=1}^{B} \sigma(\widehat{\theta}_j)$. By Assumption (i), every $\widehat{\theta}_j$ is Borel measurable, so $\sigma(\widehat{\theta}_j) \subseteq \mathcal{F}$ for every j. Therefore $\mathcal{G} \subseteq \mathcal{F}$. By Assumption (iv), the σ-algebras $\sigma(U)$ and $\mathcal{G}$ are independent; equivalently, for every $A \in \sigma(U)$ and every $C \in \mathcal{G}$,

$$P(A \cap C) = P(A)P(C).$$

We first prove that $N_k \in \mathcal{G}$. Fix n with $1 \le n \le B$. The n-th generator of $\mathcal{G}$ is $\sigma(\widehat{\theta}_n)$, hence $\sigma(\widehat{\theta}_n) \subseteq \mathcal{G}$. Since $(-\infty, \theta_0]$ and $[\theta_0, \infty)$ are Borel subsets of $\mathbb{R}$, the events

$$\{\widehat{\theta}_n \le \theta_0\}, \qquad \{\widehat{\theta}_n \ge \theta_0\}$$

both belong to $\mathcal{G}$.

Now define

$$A_- = \{\omega : \exists\, j \in \{1, \ldots, k\} \text{ with } \widehat{\theta}_j(\omega) \le \theta_0\} = \bigcup_{j=1}^{k} \{\widehat{\theta}_j \le \theta_0\},$$

and

$$A_+ = \{\omega : \exists\, j \in \{1, \ldots, k\} \text{ with } \widehat{\theta}_j(\omega) \ge \theta_0\} = \bigcup_{j=1}^{k} \{\widehat{\theta}_j \ge \theta_0\}.$$

By Assumption (iii), every index $j = 1, \ldots, k$ satisfies $j \le B$, so each set in these finite unions belongs to $\mathcal{G}$. Hence $A_- \in \mathcal{G}$ and $A_+ \in \mathcal{G}$. By the definition of N_k,

$$N_k = (A_- \cap A_+)^c,$$

so $N_k \in \mathcal{G}$. Since $\mathcal{G} \subseteq \mathcal{F}$, it follows also that $N_k \in \mathcal{F}$.

Finally, the event

$$\{U \le \tau\} = U^{-1}\big((-\infty, \tau]\big)$$

belongs to $\sigma(U)$, because U is measurable and $(-\infty, \tau]$ is Borel. Applying the independence

factorization with $A = \{U \le \tau\}$ and $C = N_k$ gives

$$P\big(\{U \le \tau\} \cap N_k\big) = P(U \le \tau)\, P(N_k).$$

Similarly,

$$\{\tau < U\} = U^{-1}\big((\tau, \infty)\big)$$

belongs to $\sigma(U)$, so applying the same independence factorization with $A = \{\tau < U\}$ and $C = N_k$ gives

$$P\big(\{\tau < U\} \cap N_k\big) = P(\tau < U)\, P(N_k).$$

Thus N_k is measurable with respect to both the ambient σ-algebra and $\mathcal{G}_B$, and its intersections with $\{U \le \tau\}$ and $\{\tau < U\}$ factor as required.

□

External node: cited lemma (AXIOM)

Lemma 2 (Finite-B coverage bound)**.** *Let $B \ge 1$ be an integer, let $(\Omega, \mathcal{F}, \mu)$ be a probability space, and let $\theta_0 \in \mathbb{R}$. For $j = 1, \dots, B$, let*

$$\widehat{\theta}_j : \Omega \to \mathbb{R}$$

be a real-valued estimator. For each j, define the median bias of $\widehat{\theta}_j$ for θ_0 by

$$\mathrm{MedBias}_{\theta_0}(\widehat{\theta}_j) := \left(\frac{1}{2} - \min\left\{\mu\big(\widehat{\theta}_j - \theta_0 \ge 0\big),\, \mu\big(\widehat{\theta}_j - \theta_0 \le 0\big)\right\}\right)_+,$$

where $(x)_+ = \max\{x, 0\}$, and put

$$\Delta := \max_{1 \le j \le B} \mathrm{MedBias}_{\theta_0}(\widehat{\theta}_j).$$

Assume:

(i) each estimator $\widehat{\theta}_j$ is measurable;

(ii) the real-valued estimators $\widehat{\theta}_1, \dots, \widehat{\theta}_B$ are independent under μ;

(iii) $\Delta \in [0, 1/2]$.

Then

$$\mu\left(\left\{\omega : \theta_0 \notin \left[\min_{1 \le j \le B} \widehat{\theta}_j(\omega),\, \max_{1 \le j \le B} \widehat{\theta}_j(\omega)\right]\right\}\right) \le \left(\frac{1}{2} - \Delta\right)^B + \left(\frac{1}{2} + \Delta\right)^B,$$

where the powers are ordinary integer powers.

This lemma is supplied as an external assumption; its proof is not part of the local Lean proof.
External assumption linked to task paper_6093c8cdedac44ebaae3246c2f49ba3b (job job_paper_6093c8cdedac44ebaae3246c2f49ba3b): being proved. No completed proof for this consumer is certified by this snapshot.

Source The cited result is the source's finite-B, finite-sample coverage lemma for independent real-valued estimators, referred to as the fixed-B coverage result. The article source is listed as "main" and the target as the fixed-B coverage lemma; no author, theorem number, or further bibliographic details were supplied. The cited dependency is the result with resultId `paper_6093c8cdedac44ebaae3246c2f49ba3b` and jobId `job_paper_6093c8cdedac44ebaae3246c2f49ba3b`.

Reference provenance Finite-B, finite-sample statement for independent real-valued estimators $\widehat{\theta}_1, \ldots, \widehat{\theta}_B$. The non-strict inequalities in the definition of median bias are important: the events are $\widehat{\theta}_j - \theta_0 \geq 0$ and $\widehat{\theta}_j - \theta_0 \leq 0$. The quantity Δ is the maximum median bias over the B estimators and lies in $[0, 1/2]$. The function $P(B; \Delta)$ is exactly $\left(\frac{1}{2} - \Delta\right)^B + \left(\frac{1}{2} + \Delta\right)^B$. The lemma gives only an overcoverage/noncoverage upper bound; it does not supply the exact-coverage equality appearing in the selected target. It is stated for general independent estimators and is not tied to the randomized choice of B^* in Algorithm 1 of the selected target. The supplied source states that the proof is relegated to the supplement, Section \ref{appsec:fixed-B-coverage-result}; the proof is not reproduced in this collection.

Use in this proof It is the external overcoverage bound used in the consumer node `fixed_batch_bound`: for a fixed number of independent estimators, it controls the probability that θ_0 falls outside the closed interval from their minimum to their maximum by $\left(\frac{1}{2} - \Delta\right)^B + \left(\frac{1}{2} + \Delta\right)^B$. It supplies only the noncoverage/overcoverage upper bound, not an exact-coverage equality.

Lemma 3 (Finite-batch noncoverage bound)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $\theta_0, \Delta \in \mathbb{R}$, let $B \geq 1$ be an integer, and let $(\widehat{\theta}_j)_{j \geq 1}$ be real-valued random variables. For*

$j \geq 1$, define the median bias

$$\mathrm{MedBias}_{\theta_0}(\widehat{\theta}_j) = \left(\frac{1}{2} - \min\left\{\mathbb{P}(\widehat{\theta}_j \geq \theta_0),\, \mathbb{P}(\widehat{\theta}_j \leq \theta_0)\right\}\right)_+,$$

where $x_+ = \max\{x, 0\}$. For integers $m \geq 1$, set

$$p_{\mathrm{mis}}(m; \Delta) = \left(\frac{1}{2} - \Delta\right)^m + \left(\frac{1}{2} + \Delta\right)^m.$$

For $k \geq 1$, write

$$\widehat{\mathrm{CI}}_k(\omega) = \left[\min_{1 \leq j \leq k} \widehat{\theta}_j(\omega),\, \max_{1 \leq j \leq k} \widehat{\theta}_j(\omega)\right], \qquad E_k = \left\{\omega : \theta_0 \notin \widehat{\mathrm{CI}}_k(\omega)\right\}.$$

Assume:

(i) $0 \leq \Delta < 1/2$;

(ii) $B \geq 1$;

(iii) each $\widehat{\theta}_j$ is measurable for $j \geq 1$;

(iv) the random variables $\widehat{\theta}_1, \ldots, \widehat{\theta}_B$ are mutually independent under $\mathbb{P}$;

(v) $\mathrm{MedBias}_{\theta_0}(\widehat{\theta}_j) \leq \Delta$ *for every* $1 \leq j \leq B$;

(vi) k is an integer with $1 \leq k \leq B$.

Then

$$\mathbb{P}(E_k) \leq p_{\mathrm{mis}}(k; \Delta).$$

Proof. Set

$$M_j = \mathrm{MedBias}_{\theta_0}(\widehat{\theta}_j) = \max\left\{\frac{1}{2} - \min\left(\mathbb{P}(\widehat{\theta}_j \geq \theta_0), \mathbb{P}(\widehat{\theta}_j \leq \theta_0)\right), 0\right\}.$$

The expression is symmetric in its two real-valued argument slots: interchanging them only interchanges the two probabilities inside the minimum. Hence the maximum-bias quantity below may be written using the same median-bias definition without ambiguity.

First observe that

$$M_j \in \left[0, \frac{1}{2}\right]$$

for every j. Indeed, both one-sided probabilities

$$a_j = \mathbb{P}(\widehat{\theta}_j \geq \theta_0), \qquad b_j = \mathbb{P}(\widehat{\theta}_j \leq \theta_0)$$

are nonnegative, so $\min(a_j, b_j) \geq 0$. Therefore

$$\frac{1}{2} - \min(a_j, b_j) \leq \frac{1}{2},$$

while the maximum with 0 gives $M_j \geq 0$. Thus $0 \leq M_j \leq 1/2$.

Define

$$\delta_k = \max_{1 \leq j \leq k} M_j.$$

Since $k \geq 1$, this is a maximum over a nonempty finite set. From $0 \leq M_j \leq 1/2$ for every j, it follows that

$$0 \leq \delta_k \leq \frac{1}{2}.$$

Also, by assumption (v) and $k \leq B$, each M_j with $1 \leq j \leq k$ satisfies $M_j \leq \Delta$. Hence

$$\delta_k \leq \Delta.$$

Next, the first k estimators are measurable, by assumption (iii), and they are mutually independent. To verify the independence, let $S \subseteq \{1, \ldots, k\}$ be finite, and for each $i \in S$ let $A_i \subseteq \mathbb{R}$ be measurable. Under the inclusion of indices into $\{1, \ldots, B\}$, the set S has the same cardinality as its image $S' \subseteq \{1, \ldots, B\}$. Define measurable sets A'_j on the B-index set by setting $A'_j = A_i$ when $j = i \in S' \subseteq \{1, \ldots, B\}$, and $A'_j = \mathbb{R}$ for indices outside S'. Applying independence of $\widehat{\theta}_1, \ldots, \widehat{\theta}_B$ to the finite set S' gives

$$\mathbb{P}\left(\bigcap_{j \in S'} \{\widehat{\theta}_j \in A'_j\}\right) = \prod_{j \in S'} \mathbb{P}(\widehat{\theta}_j \in A'_j).$$

Since the index map is injective, this is exactly

$$\mathbb{P}\left(\bigcap_{i \in S} \{\widehat{\theta}_i \in A_i\}\right) = \prod_{i \in S} \mathbb{P}(\widehat{\theta}_i \in A_i).$$

Thus the estimators $\widehat{\theta}_1, \ldots, \widehat{\theta}_k$ are mutually independent.

By the fixed-batch coverage bound, recalled as Lemma 2, applied to the independent

measurable estimators $\widehat{\theta}_1, \ldots, \widehat{\theta}_k$ and to their maximum median bias $\delta_k \in [0, 1/2]$, we have

$$\mathbb{P}\left(\theta_0 \notin \left[\min_{1\le j\le k} \widehat{\theta}_j, \max_{1\le j\le k} \widehat{\theta}_j\right]\right) \le p_{\mathrm{mis}}(k; \delta_k).$$

It remains to identify the event on the left with E_k. Because the index set $\{1, \ldots, k\}$ is nonempty, the condition

$$\theta_0 \in \left[\min_{1\le j\le k} \widehat{\theta}_j(\omega), \max_{1\le j\le k} \widehat{\theta}_j(\omega)\right]$$

is equivalent to the conjunction of the two statements

$$\exists j \in \{1, \ldots, k\},\ \widehat{\theta}_j(\omega) \le \theta_0, \qquad \exists j \in \{1, \ldots, k\},\ \theta_0 \le \widehat{\theta}_j(\omega).$$

Therefore its negation is

$$\left(\forall j \in \{1, \ldots, k\},\ \theta_0 < \widehat{\theta}_j(\omega)\right) \text{ or } \left(\forall j \in \{1, \ldots, k\},\ \widehat{\theta}_j(\omega) < \theta_0\right).$$

On the other hand, since the interval is closed,

$$\theta_0 \notin \left[\min_{1\le j\le k} \widehat{\theta}_j(\omega), \max_{1\le j\le k} \widehat{\theta}_j(\omega)\right]$$

is equivalent to

$$\theta_0 < \min_{1\le j\le k} \widehat{\theta}_j(\omega) \quad \text{or} \quad \max_{1\le j\le k} \widehat{\theta}_j(\omega) < \theta_0,$$

and these two alternatives are exactly the two universal alternatives above. Hence

$$E_k = \left\{\omega : \theta_0 \notin \left[\min_{1\le j\le k} \widehat{\theta}_j(\omega), \max_{1\le j\le k} \widehat{\theta}_j(\omega)\right]\right\}.$$

Consequently,

$$\mathbb{P}(E_k) \le p_{\mathrm{mis}}(k; \delta_k).$$

We now use monotonicity in the bias parameter. For $t \in [0, 1/2]$, define

$$F(t) = \left(\frac{1}{2} - t\right)^k + \left(\frac{1}{2} + t\right)^k.$$

On the interior $(0, 1/2)$,

$$F'(t) = k\left(\frac{1}{2} + t\right)^{k-1} - k\left(\frac{1}{2} - t\right)^{k-1} = k\left[\left(\frac{1}{2} + t\right)^{k-1} - \left(\frac{1}{2} - t\right)^{k-1}\right].$$

Since $0 \le \frac{1}{2} - t \le \frac{1}{2} + t$ and $k - 1 \ge 0$, we have

$$\left(\frac{1}{2} - t\right)^{k-1} \le \left(\frac{1}{2} + t\right)^{k-1},$$

so $F'(t) \ge 0$. Therefore F is nondecreasing on $[0, 1/2]$. Thus for all

$$0 \le a \le b \le \frac{1}{2},$$

we have

$$\left(\frac{1}{2} - a\right)^{k} + \left(\frac{1}{2} + a\right)^{k} \le \left(\frac{1}{2} - b\right)^{k} + \left(\frac{1}{2} + b\right)^{k}.$$

Applying this with $a = \delta_k$ and $b = \Delta$, using $0 \le \delta_k \le \Delta$ and $\Delta \le 1/2$, gives

$$p_{\text{mis}}(k; \delta_k) \le p_{\text{mis}}(k; \Delta).$$

Therefore

$$\mathbb{P}(E_k) \le p_{\text{mis}}(k; \Delta).$$

Finally, this is already an inequality for real-valued probabilities; since $\mathbb{P}$ is a probability measure and $p_{\text{mis}}(k; \Delta)$ is finite and nonnegative, it is the same as the asserted probability inequality. Thus

$$\mathbb{P}\left(\left\{\omega : \neg\, \theta_0 \in \widehat{\text{CI}}_k(\omega)\right\}\right) \le p_{\text{mis}}(k; \Delta),$$

which is the required bound.

□

Lemma 4 (Uniform tail probabilities)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $U : \Omega \to \mathbb{R}$ be a random variable, and let $c \in \mathbb{R}$. Let* Leb *denote Lebesgue measure on $\mathbb{R}$. Suppose:*

(i) U is measurable;

(ii) the law of U under $\mathbb{P}$ is the restriction of Lebesgue measure to $[0, 1]$, that is,

$$U_*\mathbb{P} = \text{Leb}\,|_{[0,1]};$$

(iii) $0 \le c$;

(iv) $c \le 1$.

Then

$$\mathbb{P}(U \le c) = c \qquad \textit{and} \qquad \mathbb{P}(c < U) = 1 - c.$$

Proof. For every measurable set $A \subseteq \mathbb{R}$, Condition (ii) and the measurability of U from Condition (i) give

$$\mathbb{P}(U^{-1}(A)) = (U_*\mathbb{P})(A) = \mathrm{Leb}\,|_{[0,1]}(A) = \mathrm{Leb}\big(A \cap [0,1]\big).$$

First consider the event $\{U \le c\} = U^{-1}\big((-\infty, c]\big)$. Applying the identity above to the measurable half-line $(-\infty, c]$, we obtain

$$\mathbb{P}(U \le c) = \mathrm{Leb}\big((-\infty, c] \cap [0,1]\big).$$

By Conditions (iii) and (iv), $0 \le c \le 1$, so

$$(-\infty, c] \cap [0,1] = [0, c].$$

The Lebesgue measure of $[0, c]$ is $c - 0 = c$. Hence

$$\mathbb{P}(U \le c) = c.$$

Next consider the event $\{c < U\} = U^{-1}\big((c, \infty)\big)$. Applying the same identity to the measurable half-line (c, ∞), we get

$$\mathbb{P}(c < U) = \mathrm{Leb}\big((c, \infty) \cap [0,1]\big).$$

Again using $0 \le c \le 1$, we have

$$(c, \infty) \cap [0,1] = (c, 1],$$

because every point in the intersection is greater than $c \ge 0$ and at most 1, and conversely every point greater than c and at most 1 lies in the intersection. The Lebesgue measure of $(c, 1]$ is $1 - c$. Therefore

$$\mathbb{P}(c < U) = 1 - c.$$

This proves both assertions.

□

Lemma 5 (Exact miscoverage probability for an independent median-bias-calibrated batch family)**.** *Let* $(\Omega, \mathcal{F}, \mathbb{P})$ *be a probability space. Let* $\theta_0, \Delta \in \mathbb{R}$ *satisfy*

$$0 \le \Delta < \frac{1}{2},$$

let $B \geq 1$ be an integer, and let $\widehat{\theta}_j : \Omega \to \mathbb{R}$, $j \geq 1$, be measurable random variables. Suppose the following conditions hold:

(i) The random variables $\widehat{\theta}_1, \ldots, \widehat{\theta}_B$ are independent.

(ii) For every $j = 1, \ldots, B$,

$$\mathbb{P}\{\widehat{\theta}_j \leq \theta_0\} = \mathbb{P}\{\widehat{\theta}_1 \leq \theta_0\}.$$

(iii) For every $j = 1, \ldots, B$,

$$\mathbb{P}\{\widehat{\theta}_j = \theta_0\} = 0.$$

(iv) For every $j = 1, \ldots, B$,

$$\mathrm{MB}_{\theta_0}(\widehat{\theta}_j) = \Delta,$$

where for a real random variable Y,

$$\mathrm{MB}_{\theta_0}(Y) = \max\left\{\frac{1}{2} - \min\big(\mathbb{P}\{Y \geq \theta_0\}, \mathbb{P}\{Y \leq \theta_0\}\big), 0\right\}.$$

For an integer k with $1 \leq k \leq B$, define

$$\mathrm{CI}_k(\omega) = \left[\min_{1\leq j\leq k} \widehat{\theta}_j(\omega), \max_{1\leq j\leq k} \widehat{\theta}_j(\omega)\right].$$

Equivalently, $\theta_0 \in \mathrm{CI}_k(\omega)$ if and only if there exist $j, j' \in \{1, \ldots, k\}$ such that

$$\widehat{\theta}_j(\omega) \leq \theta_0 \leq \widehat{\theta}_{j'}(\omega).$$

Then

$$\mathbb{P}\big\{\omega : \theta_0 \notin \mathrm{CI}_k(\omega)\big\} = p_\Delta(k),$$

where

$$p_\Delta(k) = \left(\frac{1}{2} - \Delta\right)^k + \left(\frac{1}{2} + \Delta\right)^k.$$

Proof. Set

$$a = \mathbb{P}\{\theta_0 \leq \widehat{\theta}_1\}, \qquad b = \mathbb{P}\{\widehat{\theta}_1 \leq \theta_0\}.$$

These are probabilities, so $a, b \geq 0$, and since the underlying measure is a probability measure, they are real numbers in $[0, 1]$.

We first record a useful consequence of the zero-atom condition. Let Y be a measurable real random variable with

$$\mathbb{P}\{Y = \theta_0\} = 0.$$

Then

$$\mathbb{P}\{\theta_0 \le Y\} = \mathbb{P}\{\theta_0 < Y\} \qquad \text{and} \qquad \mathbb{P}\{Y \le \theta_0\} = \mathbb{P}\{Y < \theta_0\}. \tag{1}$$

Indeed, for the first identity,

$$\{\theta_0 < Y\} \subseteq \{\theta_0 \le Y\} \subseteq \{\theta_0 < Y\} \cup \{Y = \theta_0\}.$$

The last union has probability at most

$$\mathbb{P}\{\theta_0 < Y\} + \mathbb{P}\{Y = \theta_0\} = \mathbb{P}\{\theta_0 < Y\},$$

so the three probabilities are equal. The second identity follows in the same way from

$$\{Y < \theta_0\} \subseteq \{Y \le \theta_0\} \subseteq \{Y < \theta_0\} \cup \{Y = \theta_0\}.$$

Also, for any measurable Y,

$$\{\theta_0 < Y\} = \{Y \le \theta_0\}^c.$$

Hence

$$\mathbb{P}\{\theta_0 < Y\} + \mathbb{P}\{Y \le \theta_0\} = 1. \tag{2}$$

Here the two probabilities are identified with their real values; since the underlying measure is finite, addition in the extended nonnegative sense agrees with ordinary addition.

Apply this to $Y = \widehat{\theta}_1$. Condition (iii) gives

$$\mathbb{P}\{\widehat{\theta}_1 = \theta_0\} = 0.$$

By (1),

$$\mathbb{P}\{\theta_0 < \widehat{\theta}_1\} = \mathbb{P}\{\theta_0 \le \widehat{\theta}_1\} = a.$$

Using (2) with $Y = \widehat{\theta}_1$, we obtain

$$a + b = 1. \tag{3}$$

Now fix $j \in \{1, \ldots, B\}$. Condition (ii) gives

$$\mathbb{P}\{\widehat{\theta}_j \le \theta_0\} = b.$$

Condition (iii) says that the atom at θ_0 has probability zero. Applying (2) to $\widehat{\theta}_j$, we get

$$\mathbb{P}\{\theta_0 < \widehat{\theta}_j\} = a.$$

Applying (1) to $\widehat{\theta}_j$, we also get

$$\mathbb{P}\{\widehat{\theta}_j < \theta_0\} = \mathbb{P}\{\widehat{\theta}_j \leq \theta_0\} = b.$$

Thus, for every $j = 1, \ldots, B$,

$$\mathbb{P}\{\theta_0 < \widehat{\theta}_j\} = a, \qquad \mathbb{P}\{\widehat{\theta}_j < \theta_0\} = b. \tag{4}$$

Using condition (iv) for $j = 1$ and the definition of the median bias, we have

$$\Delta = \max\left\{\frac{1}{2} - \min(a, b),\, 0\right\}.$$

Since $a + b = 1$ by (3), we have $\min(a, b) \leq 1/2$, so

$$\frac{1}{2} - \min(a, b) \geq 0.$$

Therefore the maximum is the first entry, and

$$\Delta = \frac{1}{2} - \min(a, b).$$

Equivalently,

$$\min(a, b) = \frac{1}{2} - \Delta. \tag{5}$$

We next identify a and b up to order. If $b \leq 1/2$, then $b \leq a$ by (3), hence $\min(a, b) = b$. From (5),

$$b = \frac{1}{2} - \Delta, \qquad a = \frac{1}{2} + \Delta.$$

If instead $b > 1/2$, then $a \leq b$ by (3), hence $\min(a, b) = a$. From (5),

$$a = \frac{1}{2} - \Delta, \qquad b = \frac{1}{2} + \Delta.$$

In either case,

$$a^k + b^k = \left(\frac{1}{2} - \Delta\right)^k + \left(\frac{1}{2} + \Delta\right)^k = p_\Delta(k). \tag{6}$$

Now define the two events

$$E_+ = \big\{\omega : \forall j \in \{1, \ldots, k\},\ \theta_0 < \widehat{\theta}_j(\omega)\big\},$$

$$E_- = \big\{\omega : \forall j \in \{1, \ldots, k\},\ \widehat{\theta}_j(\omega) < \theta_0\big\}.$$

Equivalently,

$$E_+ = \bigcap_{j=1}^{k} \{\theta_0 < \widehat{\theta}_j\}, \qquad E_- = \bigcap_{j=1}^{k} \{\widehat{\theta}_j < \theta_0\}.$$

Since each $\widehat{\theta}_j$ is measurable, each of these finite intersections is measurable. Because $k \geq 1$, the index set $\{1, \ldots, k\}$ is nonempty. If $\omega \in E_+ \cap E_-$, then choosing any $j \in \{1, \ldots, k\}$ would give both $\theta_0 < \widehat{\theta}_j(\omega)$ and $\widehat{\theta}_j(\omega) < \theta_0$, which is impossible. Hence

$$E_+ \cap E_- = \varnothing.$$

We claim that the noncoverage event is exactly $E_+ \cup E_-$:

$$\big\{\omega : \theta_0 \notin \mathrm{CI}_k(\omega)\big\} = E_+ \cup E_-. \tag{7}$$

Indeed, $\theta_0 \in \mathrm{CI}_k(\omega)$ means that some estimator lies at or below θ_0 and some estimator lies at or above θ_0. If $\omega \notin \mathrm{CI}_k(\omega)$ and not all $\widehat{\theta}_j(\omega)$ are strictly greater than θ_0, then some j satisfies $\widehat{\theta}_j(\omega) \leq \theta_0$. If also some j' satisfied $\theta_0 \leq \widehat{\theta}_{j'}(\omega)$, then θ_0 would lie between the minimum and maximum, contradicting noncoverage. Therefore all estimators are strictly below θ_0, so $\omega \in E_-$. Thus noncoverage implies either E_+ or E_-. Conversely, if all estimators are strictly above θ_0, then no estimator is at or below θ_0, so θ_0 is not in the interval. If all estimators are strictly below θ_0, then no estimator is at or above θ_0, so again θ_0 is not in the interval. This proves (7).

Now let

$$S = \{1, \ldots, k\} \subseteq \{1, \ldots, B\}.$$

The set S has cardinality k, and

$$E_+ = \bigcap_{j \in S} \{\theta_0 < \widehat{\theta}_j\}, \qquad E_- = \bigcap_{j \in S} \{\widehat{\theta}_j < \theta_0\}.$$

By independence of $\widehat{\theta}_1, \ldots, \widehat{\theta}_B$, condition (i), the probability of an intersection over any subset of these variables factors as the product of the individual probabilities. Therefore,

using (4),

$$\mathbb{P}(E_+) = \prod_{j \in S} \mathbb{P}\{\theta_0 < \widehat{\theta}_j\} = \prod_{j=1}^{k} a = a^k,$$

and similarly

$$\mathbb{P}(E_-) = \prod_{j \in S} \mathbb{P}\{\widehat{\theta}_j < \theta_0\} = \prod_{j=1}^{k} b = b^k.$$

The product of constants is converted to the corresponding power using $a, b \geq 0$.

Finally, since E_+ and E_- are disjoint measurable events, the additivity of probability gives

$$\mathbb{P}\big\{\omega : \theta_0 \notin \mathrm{CI}_k(\omega)\big\} = \mathbb{P}(E_+ \cup E_-) = \mathbb{P}(E_+) + \mathbb{P}(E_-) = a^k + b^k.$$

By (6), this equals $p_\Delta(k)$, namely

$$\left(\frac{1}{2} - \Delta\right)^k + \left(\frac{1}{2} + \Delta\right)^k.$$

Thus

$$\mathbb{P}\big\{\omega : \theta_0 \notin \mathrm{CI}_k(\omega)\big\} = p_\Delta(k),$$

as claimed.

□

Main Theorem

Theorem 6 (HulC finite-sample coverage and exact coverage)**.** *Let $n \geq 1$, $0 < \alpha < 1$, $0 \leq \Delta < 1/2$, and $\theta_0 \in \mathbb{R}$. For integers $k \geq 0$, define*

$$\mathcal{P}(k; \Delta) = \left(\frac{1}{2} - \Delta\right)^k + \left(\frac{1}{2} + \Delta\right)^k.$$

Let $B_{\alpha,\Delta} \geq 1$ be the smallest integer B such that $\mathcal{P}(B; \Delta) \leq \alpha$, and write $B = B_{\alpha,\Delta}$. Define

$$\tau_{\alpha,\Delta} = \frac{\alpha - \mathcal{P}(B; \Delta)}{\mathcal{P}(B-1; \Delta) - \mathcal{P}(B; \Delta)}.$$

For a random variable $U : \Omega \to \mathbb{R}$, put

$$B^*(\omega) = \begin{cases} B, & \tau_{\alpha,\Delta} = 0, \\ B-1, & \tau_{\alpha,\Delta} \neq 0 \text{ and } U(\omega) \leq \tau_{\alpha,\Delta}, \\ B, & \tau_{\alpha,\Delta} \neq 0 \text{ and } U(\omega) > \tau_{\alpha,\Delta}. \end{cases}$$

Let $(\Omega, \mathcal{F}, P)$ be a probability space, let $(\mathcal{X}, \mathcal{A})$ be a measurable space, let $X_1, \ldots, X_n : \Omega \to \mathcal{X}$, let $S_1, S_2, \cdots : \Omega \to \mathcal{P}(\{1, \ldots, n\})$, and let $\widehat{\theta}_1, \widehat{\theta}_2, \cdots : \Omega \to \mathbb{R}$. For an estimator $\widehat{\theta}$, define its median bias at θ_0 by

$$\mathrm{MedBias}_{\theta_0}(\widehat{\theta}) = \left(\frac{1}{2} - \min\{\mathbb{P}(\widehat{\theta} \geq \theta_0), \mathbb{P}(\widehat{\theta} \leq \theta_0)\} \right)_+,$$

where $(x)_+ = \max\{x, 0\}$. For $k \geq 0$, let

$$E_k = \left\{ \omega : \left(\exists\, j \in \{1, \ldots, k\} : \widehat{\theta}_j(\omega) \leq \theta_0\right) \text{ and } \left(\exists\, j \in \{1, \ldots, k\} : \theta_0 \leq \widehat{\theta}_j(\omega)\right) \right\},$$

with the empty existential understood when $k = 0$, and set

$$E^* = \left\{ \omega : \omega \in E_{B^*(\omega)} \right\}.$$

For $k \geq 1$, E_k is equivalently the event that $\theta_0 \in [\min_{1 \leq j \leq k} \widehat{\theta}_j, \max_{1 \leq j \leq k} \widehat{\theta}_j]$. Assume:

(i) $B_{\alpha,\Delta} \leq n$.

(ii) $X_1, \ldots, X_n$ *are independent and measurable.*

(iii) Each $\widehat{\theta}_j$ *is measurable, and* $\widehat{\theta}_1, \ldots, \widehat{\theta}_B$ *are independent.*

(iv) $\mathrm{MedBias}_{\theta_0}(\widehat{\theta}_j) \leq \Delta$ *for* $j = 1, \ldots, B$.

(v) U *is measurable and* $U \sim \mathrm{Unif}[0, 1]$.

(vi) U *is independent of the sigma-algebra generated by* $X_1, \ldots, X_n$.

(vii) U *is independent of the sigma-algebra generated by* $\widehat{\theta}_1, \ldots, \widehat{\theta}_B$.

(viii) U *is independent of the sigma-algebra generated by* $S_1, S_2, \ldots$.

(ix) For every $j \geq 1$, $\widehat{\theta}_j$ *is measurable with respect to the sigma-algebra generated by* $X_1, \ldots, X_n$ *and* S_j.

(x) For every ω, the sets $S_j(\omega)$, $1 \le j \le B^*(\omega)$, are pairwise disjoint and their union is $\{1, \dots, n\}$.

Then

$$\mathbb{P}(E^*) \ge 1 - \alpha.$$

Moreover, if in addition

(xi) $\mathbb{P}(\widehat{\theta}_j \le \theta_0) = \mathbb{P}(\widehat{\theta}_1 \le \theta_0)$ *for* $j = 1, \dots, B$;

(xii) $\mathbb{P}(\widehat{\theta}_j = \theta_0) = 0$ *for* $j = 1, \dots, B$;

(xiii) $\mathrm{MedBias}_{\theta_0}(\widehat{\theta}_j) = \Delta$ *for* $j = 1, \dots, B$,

then

$$\mathbb{P}(E^*) = 1 - \alpha.$$

Proof. Write

$$p_k = \mathcal{P}(k; \Delta) = \left(\frac{1}{2} - \Delta\right)^k + \left(\frac{1}{2} + \Delta\right)^k, \qquad B = B_{\alpha,\Delta}, \qquad \tau = \tau_{\alpha,\Delta}.$$

Since $0 \le \Delta < 1/2$, both

$$a = \frac{1}{2} - \Delta > 0, \qquad b = \frac{1}{2} + \Delta > 0$$

are positive. Hence $p_k > 0$ for every k.

First, $p_1 = a + b = 1$. Since $\alpha < 1$, we have $p_1 > \alpha$. If $B = 1$, then the defining property $p_B \le \alpha$ would give $p_1 \le \alpha$, a contradiction. Thus $B \ne 1$, and since $B \ge 1$, we get $B \ge 2$. Consequently $B - 1 \ge 1$ and $B - 1 \le B$.

For every $k \ge 0$,

$$p_k - p_{k+1} = a^k b + b^k a > 0,$$

so

$$p_{k+1} < p_k.$$

Thus $k \mapsto p_k$ is strictly decreasing. Applying minimality to $B - 1$, which is an admissible positive integer, shows that $p_{B-1} \le \alpha$ is impossible; hence

$$p_{B-1} > \alpha.$$

Together with $p_B \le \alpha$, this gives

$$p_{B-1} - p_B > 0.$$

Therefore the denominator in the definition of τ is positive, while the numerator $\alpha - p_B$ is nonnegative. It follows that

$$0 \le \tau \le 1,$$

where $\tau \le 1$ is equivalent to $\alpha \le p_{B-1}$. From the definition of τ and the positivity of $p_{B-1} - p_B$, we also get the mixing identity

$$\tau p_{B-1} + (1-\tau)p_B = \alpha.$$

Moreover,

$$\tau = 0 \quad \Longleftrightarrow \quad \alpha = p_B.$$

Let

$$A_0 = \{U \le \tau\}, \qquad A_1 = \{\tau < U\}.$$

Both are measurable because U is measurable. For $k \ge 1$, let

$$N_k = \Omega \setminus E_k$$

be the noncoverage event for the deterministic batch count k. Equivalently, for $k \ge 1$, the event N_k says that either all $\widehat{\theta}_j > \theta_0$ for $1 \le j \le k$, or all $\widehat{\theta}_j < \theta_0$ for $1 \le j \le k$.

By the independence of U from the sigma-algebra generated by the estimators, Lemma 1 gives the measurability of N_{B-1} and N_B, together with the factorization identities

$$\mathbb{P}(A_0 \cap N_{B-1}) = \mathbb{P}(A_0)\mathbb{P}(N_{B-1}), \qquad \mathbb{P}(A_1 \cap N_B) = \mathbb{P}(A_1)\mathbb{P}(N_B).$$

By the fixed-batch coverage bound, Lemma 3, using the median-bias bound and the independence of the first B estimators,

$$\mathbb{P}(N_{B-1}) \le p_{B-1}, \qquad \mathbb{P}(N_B) \le p_B.$$

By the uniform-tail computation, Lemma 4, since $0 \le \tau \le 1$,

$$\mathbb{P}(A_0) = \tau, \qquad \mathbb{P}(A_1) = 1 - \tau.$$

The events A_0 and A_1 are disjoint, so $A_0 \cap N_{B-1}$ and $A_1 \cap N_B$ are disjoint. If $\tau = 0$, then by definition $B^* = B$ everywhere, and hence

$$N_{B^*} = N_B.$$

If $\tau \neq 0$, then pointwise $B^* = B - 1$ on A_0 and $B^* = B$ on A_1. Therefore

$$N_{B^*} = (A_0 \cap N_{B-1}) \cup (A_1 \cap N_B).$$

This identity also shows that N_{B^*} is measurable: in the case $\tau = 0$ it is N_B, and in the case $\tau \neq 0$ it is a finite union of measurable intersections.

We now estimate $\mathbb{P}(N_{B^*})$. If $\tau = 0$, then $B^* = B$, and $\tau = 0$ gives $\alpha = p_B$. Hence

$$\mathbb{P}(N_{B^*}) = \mathbb{P}(N_B) \leq p_B = \alpha.$$

If $\tau \neq 0$, then the decomposition, disjointness, factorization, uniform tails, the fixed-batch bounds, and the mixing identity give

$$\begin{aligned}
\mathbb{P}(N_{B^*}) &= \mathbb{P}(A_0 \cap N_{B-1}) + \mathbb{P}(A_1 \cap N_B) \\
&= \mathbb{P}(A_0)\mathbb{P}(N_{B-1}) + \mathbb{P}(A_1)\mathbb{P}(N_B) \\
&= \tau\, \mathbb{P}(N_{B-1}) + (1 - \tau)\, \mathbb{P}(N_B) \\
&\leq \tau p_{B-1} + (1 - \tau) p_B \\
&= \alpha.
\end{aligned}$$

Thus in all cases

$$\mathbb{P}(N_{B^*}) \leq \alpha.$$

Since E^* is the complement of N_{B^*}, and N_{B^*} is measurable with probability at most 1,

$$\mathbb{P}(E^*) = 1 - \mathbb{P}(N_{B^*}) \geq 1 - \alpha.$$

This proves the first assertion.

For the second assertion, assume the additional conditions (xi), (xii), and (xiii). If $\tau = 0$, then $B^* = B$ and $\alpha = p_B$. Applying the exact fixed-batch coverage result, Lemma 5, to the deterministic batch count B, using (xi)–(xiii), yields

$$\mathbb{P}(N_B) = p_B.$$

Hence $\mathbb{P}(N_{B^*}) = \alpha$.

If $\tau \neq 0$, Lemma 5 applied to the deterministic batch counts $B - 1$ and B, using the same additional conditions, gives

$$\mathbb{P}(N_{B-1}) = p_{B-1}, \qquad \mathbb{P}(N_B) = p_B.$$

The same decomposition and factorization as above therefore yield

$$\mathbb{P}(N_{B^*}) = \tau\,\mathbb{P}(N_{B-1}) + (1-\tau)\,\mathbb{P}(N_B) = \tau p_{B-1} + (1-\tau)p_B = \alpha.$$

Thus, in either case,

$$\mathbb{P}(N_{B^*}) = \alpha.$$

Taking complements again,

$$\mathbb{P}(E^*) = 1 - \mathbb{P}(N_{B^*}) = 1 - \alpha.$$

This proves the exact-coverage assertion.

□

Task outcomes and dependencies

Task registry snapshot: paper_fresh_20261030t143835550708z_hulc, revision 21, captured 2026-10-30T16:03:32.184Z.
This report belongs to version: v1 theorem_set: document-goal source_tag: document-processing theorems: - id: entry:selected:799d1e73-1148-4e2e-8483-d4bf5058b6cf title: Theorem 1 (thm:coverage-algorithm-known-Delta; HulC finite-sample coverage and exact coverage under the stated additional conditions) kind: theorem latex_statement: |- Suppose \(\theta_0\in\mathbb{R}\) is a parameter or functional of interest. Let \(X_1, \ldots, X_n\) be independent random variables from some measurable space \(\mathcal{X}\). For \(B\g [paper_fresh_20261030t143835550708z_hulc] (job paper_fresh_20261030t143835550708z_hulc); its own terminal report determines its outcome, since registry settlement can lag reporting.
Lemma \ref{lem:fixed-B-coverage-result} [paper_6093c8cdedac44ebaae3246c2f49ba3b] (job job_paper_6093c8cdedac44ebaae3246c2f49ba3b): running; proving; dependencies: none.
External assumption linked to task paper_6093c8cdedac44ebaae3246c2f49ba3b (job job_paper_6093c8cdedac44ebaae3246c2f49ba3b): being proved. No completed proof for this consumer is certified by this snapshot.
Each task has its own outcome. A proved child does not certify the main task or discharge other external assumptions.